

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

OPTICAL COMMUNICATION AND NETWORKING

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Define the principle of light transmission in a fiber	M 1.01	U
2	For $n_1 = 1.55$ and $n_2 = 1.52$. calculate the critical angle and numerical aperture	M 1.02	A
3	List two features of optical sources	M 2.01	R
4	List two applications of Optical fiber	M 1.03	R
5	Draw the structure of a LASER Diode	M 2.02	U
6	State expanded form of EDFA	M 3.01	R
7	State Population Inversion.	M 3.02	R
8	Define SONET	M 4.04	U
9	List any two types of fiber coupler	M 4.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	Explain parameters of Optical fiber	M 1.02	U
2	Explain any three advantages of Optical fibre	M 1.03	U
3	Light travels through water with a speed of 2.25×10^8 m/s. What is the refractive index of water? Speed of light in vacuum is 3×10^8 m/s.	M 1.01	A
4	An LED has radiative and non- radiative recombination times of 30ns and 100 ns respectively. Determine the internal quantum efficiency.	M 2.02	U
5	Explain the working of PIN photodiode	M 2.04	U

6	Explain Multiplexing Techniques used in Fiber Optic Communication system	M 3.03	U
7	Differentiate RAMAN optical amplifier and SOA	M 3.01	U
8	Differentiate Intra and Inter mode dispersion losses	M 3.02	U
9	State the function of splicer in optical fibres.	M 4.01	U
10	Define Optical Isolator, Coupler, Splitter	M 4.02	R

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Compare Surface Emitting and Edge emitting LEDs	M2.02	U
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OR

2	Explain stimulated emission of radiation in a laser diode	M 2.01	U
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3	Explain the principle of light transmission in a fibre with the help of a diagram.	M 1.01	U
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OR

4	Light travels from rarer medium 1 to a denser medium 2. The angle of incident and refraction are respectively 45° and 30°. Calculate the refractive index of second medium with respect to the first medium	M 1.02	U
5	Explain the working of a directional coupler	M 4.02	U

OR

6	Compare SDH and SONET	M 4.04	U
7	Explain the principle of photo detection	M 2.03	U

OR

8	Compare and contrast PIN and Avalanche photodiodes	M 2.04	U
9	Explain different types of losses in Optical fibers.	M 3.02	U

OR

10	Illustrate the working of Optical Receiver with block diagram.	M 3.03	U
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11	Compare Broadcast-and-select Networks and Wavelength-Routed Networks	M 4.04	U
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OR

12	Outline the Basic concepts in optical networking	M 4.03	R
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